

Week 6 Graded Mini Project: Large Language Models & Embeddings

Introduction

This **Week 20 Graded Mini Project** asks you to build a **Flowise-powered jewellery retail chatbot** for a fictional brand, **Ornativa Jewels** (Hyderabad, India). The chatbot must answer customer questions by reading and using the product catalogue provided in **Jewellery Details.pdf**. The goal is to practise building a retrieval-based chatbot using a **visual pipeline** (document loading, chunking, embeddings, retrieval, prompting, memory, and testing) without relying on hard-coded answers.

Where to find the tasks

The complete brief and the provided catalogue file ([Jewellery Details.pdf](#)) are available under **Week 20 – Graded Mini Project** on Vocareum. Follow the steps below to build your chatbot in Flowise and prepare your submission.

Note: This assignment must be completed on **Flowise** (cloud or local). It **cannot** be completed inside the **Vocareum** notebook environment.

Project Objective

Your objective is to create and test a **Jewellery Chatbot** that:

- answers customer questions using the catalogue content provided in **Jewellery Details.pdf**,
- supports follow-up questions using short-term conversation context (memory),
- remains polite, concise, and factual in tone,
- and suggests a relevant alternative when an item is out of stock (based on the catalogue).

Tasks

Task 1 – Data Setup

1. Upload the provided file **Jewellery Details.pdf** into Flowise.
2. Use a **Document Loader** node to read the contents of the PDF.

3. Connect it to a **Text Splitter** node (e.g., RecursiveCharacterTextSplitter) to chunk the content.
4. Store embeddings using a **Vector Store** node (e.g., Chroma, FAISS, or Supabase).

Deliverable: A Flowise flow that successfully ingests and indexes the jewellery catalogue.

Task 2 – Create the Query Chain

1. Add a **Chat Model** node (e.g., OpenAI GPT-4o-mini).
2. Create a **Prompt Template** node with the instruction:
“You are a jewellery store assistant. Use the catalogue data to answer questions accurately.”
3. Connect **Prompt Template** → **Retriever** → **Chat Model** to form a retrieval-based pipeline.
4. Add **Chat Input** and **Chat Output** nodes to complete the conversation flow.

Deliverable: A working query-response flow that can answer questions such as:

- “List all available diamond items.”
- “What is the price of the Pearl Necklace?”
- “Which products are made of 22K gold?”

Task 3 – Add Conversational Memory

1. Insert a **Memory** node (e.g., Buffer Memory) between the input and model nodes.
2. Ensure the chatbot can remember the previous query context. Example:

User: Show me diamond products.

Bot: Classic Diamond Ring, Diamond Stud Earrings, and Lotus Pendant are available.

User: What’s the price of the ring?

3. Test multiple turns of conversation to confirm memory retention.

Deliverable: A Flowise chatbot that maintains short-term conversation context.

Task 4 – Customisation and Testing

1. Customise the system prompt to reflect the brand’s tone, e.g.:
“You are Ornativa’s virtual jewellery expert. Be polite, concise, and factual.”
2. Test at least **five** queries, including price, material, category, and stock-related questions.
3. If an item is out of stock, configure the chatbot to suggest an alternative product. Example:
“Ruby Solitaire Ring is currently out of stock. You may like the Classic Diamond Ring instead.”

Deliverable: Evidence of testing (queries + responses) and out-of-stock handling with an appropriate alternative suggestion (based on the catalogue).

Deliverables

Mandatory

- **Flowise chatflow export** (JSON), or a clearly captured set of screenshots showing the full node pipeline and connections.
- **Test evidence:** at least **five** user queries with the chatbot’s responses (screenshots or copied logs).
- **Short write-up** (README or PDF) describing: your node choices, chunking approach, prompt design, memory setup, and how you handled out-of-stock suggestions.

Constraints

- Do not hard-code answers.
- Do not ignore the provided catalogue; responses must be driven by retrieval from **Jewellery Details.pdf**.
- Do not paste API keys into screenshots or shared files.

Practice Guidelines (for Graded Mini Project):

This graded mini project will not be completed in Vocareum. You will build a **retrieval-based conversational assistant** using **Flowise** with an LLM provider (for example,

the **OpenAI API**), focusing on document ingestion, retrieval quality, grounded prompting, memory, and testing.

To Use Chat Support for This Assignment (inside Vocareum):

- Open your practice notebook in **Vocareum**. Use the left sidebar to open the **Assistant** panel.
- Ask for clarifications, troubleshooting help, or prompt-writing support in the **Assistant** chat.
- Monitor your **GenAI Allowance/Budget**, as shown in the **Assistant panel**, and plan prompts accordingly.
- If you exhaust your allowance for a task, please contact the Emeritus support team.
- **API key access:** Your course-specific API key and usage notes are provided via the guide and Vocareum instructions. Follow those steps to locate your key when needed.
- You can find detailed instructions here: [Vocareum GenAI API Students Guide.pdf](#)

Security warning: Keep your API key *private*. Do not paste it in screenshots, chats, notebooks you share, or public repos. Prefer environment variables or secure key fields in tools; never hard-code keys in shared files. If you suspect exposure, rotate the key immediately per the guide.

To access Vocareum:

- Go to the *Getting Started* module on your Course homepage (just below Programme Orientation).
- Click the link labelled *Vocareum: Professional Certificate Programme in Agentic AI and Applications*.
- You can also access it from the Course menu on the left side of your screen.

Note: You may use Vocareum to access course guidance and API key instructions if needed. However, the chatbot build for this graded mini project must be completed in **Flowise** and cannot be executed inside Vocareum.

Submission Guidelines

Submit **one zip file** containing the following:

- **Flowise export** (JSON) or full workflow screenshots.
- **Test evidence** (at least five queries with responses).

- **README** (or short PDF) explaining your flow, prompt, memory choice, and out-of-stock handling approach.

Name your file as: **Week 20_Graded Mini Project_[Your Name].zip** (or follow the naming convention shown in the LMS if different).

Note: *This is a required graded activity and counts towards programme completion. Ensure your chatbot retrieves from the provided catalogue, maintains short-term context where required, and includes clear testing evidence. Your submission should make your design choices easy to evaluate.*